



POLITECNICO
MILANO 1863

In-core instrumentation: “Power distribution reconstruction from in-core detector measurements using gaussian process regression” Silva, A. S. et al. (2025)



In-core instrumentation

01

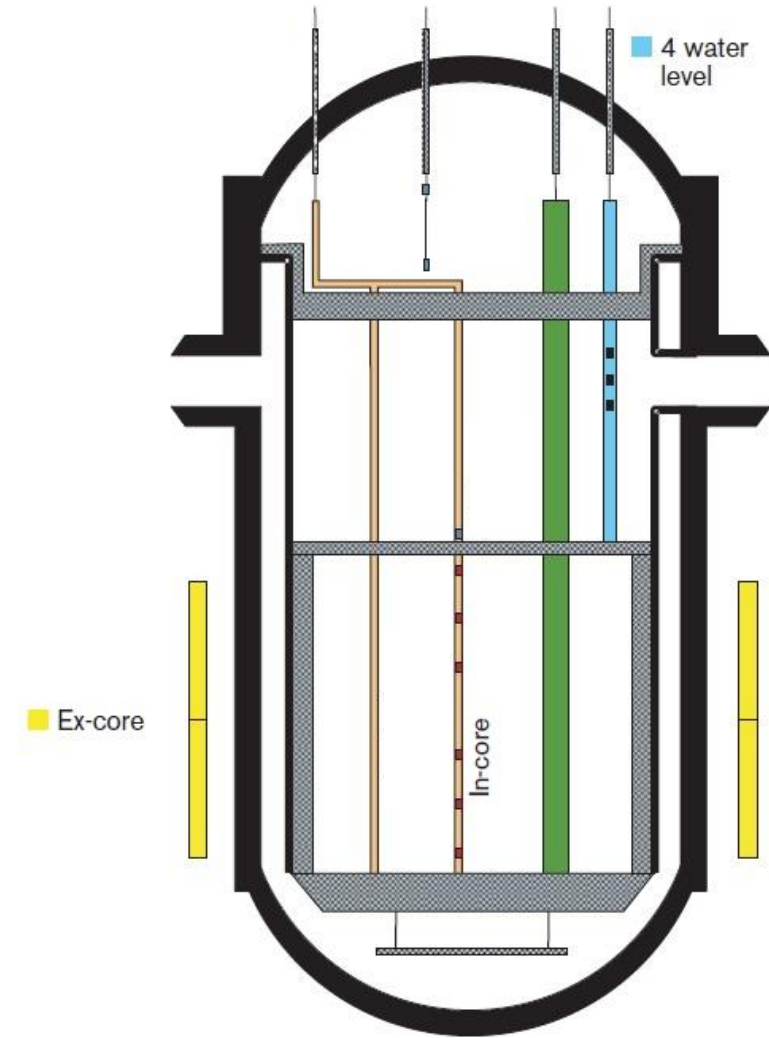
In-core instrumentation

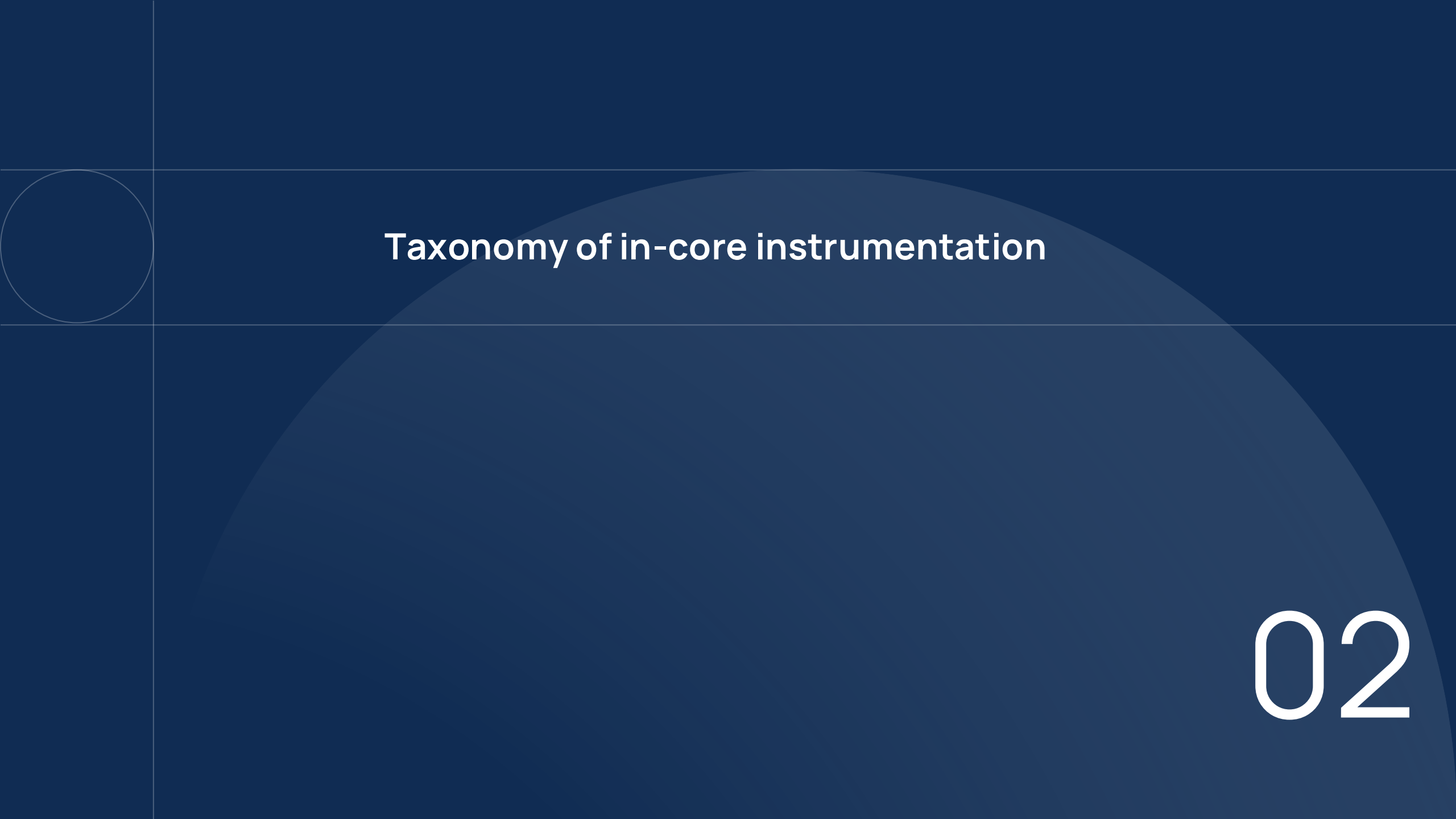
PWR instrumentation for the monitoring of neutron fluence can be **ex-core** (leaking neutrons from outside the RV) or **in-core** (detectors inserted into the RV)

1. detectors measure **local neutron fluence** (or flux) at discrete in-core locations;
2. using reactor physics models, neutron fluence is converted into **local fission rate**;
3. fission-rate distribution is reconstructed into a **3D power map**, consistent with measures of total thermal power and core calculations

Allows safe and efficient operation of PWRs by

- verification of control-rod alignment;
- detection of axial and radial power asymmetries





Taxonomy of in-core instrumentation

02

Taxonomy of in-core instrumentation

In-core instrumentation can be classified by:

- **mobility:** **fixed in-core detectors** provide continuous signals at a limited set of locations / **movable systems** perform periodic scans (*flux mapping*) with fission chamber moving axially;
- **measured field:** **nuclear** signals (flux, gamma-related) / **thermal** signals (thermocouples);
- **transduction principle:** **self-powered** detectors (SPND) / **externally biased** detectors (e.g. fission chambers);
- **time response:** **prompt-response** sensors / **delayed-response** (SPNDs);
- **spatial coverage:** «**sparse but continuous**» (only at instrumented assemblies but uninterrupted signal) / «**dense but campaign-based**» (high-resolution scanning but only during measurement campaigns);
- **function** in core monitoring: **continuous trending** (fixed) / high-resolution **calibration of 3D power maps** (movable flux mapping)



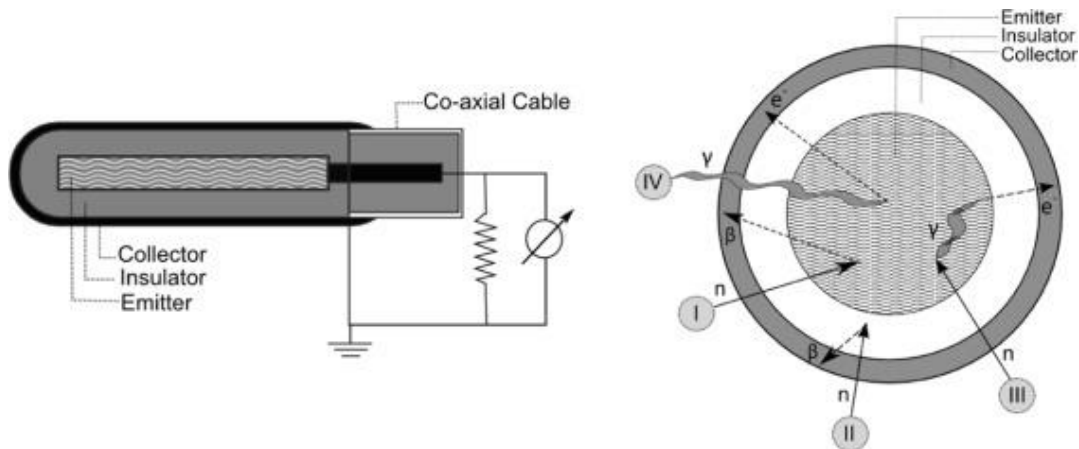
Fixed vs. movable in-core detectors

03

Fixed vs. movable in-core detectors

Fixed detectors (Self Powered Neutron Detectors, SPNDs)

- permanently installed (typically in selected instrumented assemblies/axial levels), provide **continuous trending signals** but at sparse spatial locations;
- neutrons activate emitter (Rh or V), the subsequent **beta processes generate a small current** between emitter and collector so no external bias is needed;
- current is **proportional to local neutron flux**;
- **delayed response** due to activation and decay time constants - > less suited for fast transients



Movable detectors (U_3O_8 fission chambers)

- **miniature fission chambers** containing **uranium oxide** as fissile deposit;
- fission fragments **ionize** the chamber gas and produce measurable signal;
- each detector is attached to a flexible cable and inserted into dedicated **instrumentation tubes**;
- detector is driven to the bottom of the fuel assembly, then pulled back at constant speed: **multiple axial readings** are acquired **during withdrawal** and summed;
- full-core maps require **reconstruction/interpolation** for non-instrumented assemblies





Power distribution reconstruction

04

Power distribution reconstruction

3D power distribution reconstruction is essential to **evaluate key operational parameters** that define thermal-hydraulic safety margins e.g. **enthalpy rise factor** (measures how uneven power is distributed), departure from nucleate boiling ratio (**DNBR**, CHF/local HF ratio) or **quadrant tilt ratio** (compares power among core quadrants)

in-core instrumentation provides assembly-wise, axially integrated neutron flux only at a **limited number of instrumented fuel assemblies** (to reduce cost and complexity)



power in the remaining assemblies must be **estimated** from sparse measurements

Supervised learning problem: unknown function (core power distribution) is inferred via limited measured data

Gaussian Process Regression methodological approach

- **non-parametric**: does not assume a fixed functional form and estimate a finite set of parameters, instead infers the functional itself;
- **Bayesian**, power distribution described by a probability distribution



Dataset and methods

05

Dataset and methods

- due to the lack of full-core experimental power maps, the researchers **emulated** data from the **Angra 1** nuclear power plant, a two-loop **PWR** in Brazil;
- assembly-wise integrated power values for all **121 fuel assemblies** across multiple depletion steps;
- data from the **36 instrumented assemblies** served as the training set, while data from the remaining **85 non-instrumented assemblies** was used as the reference (test) set to validate the model's accuracy
- **OpenMC** Monte Carlo code to create a fully heterogeneous **core model**, implemented in Python using the GPflow library;
- **GPR** predicts a probability distribution (mean and variance) for every point, allowing it to estimate its own uncertainty: $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$;
- transformed the core's square lattice into **centralized polar coordinates** (r, θ) to better exploit the reactor's natural symmetry;
- they utilized a custom **kernel**, a mathematical function that measures similarity between points, combining a rational **quadratic** kernel for radial distance and a **periodic** kernel for angular positions



Results and interpretation

06

Results and interpretation

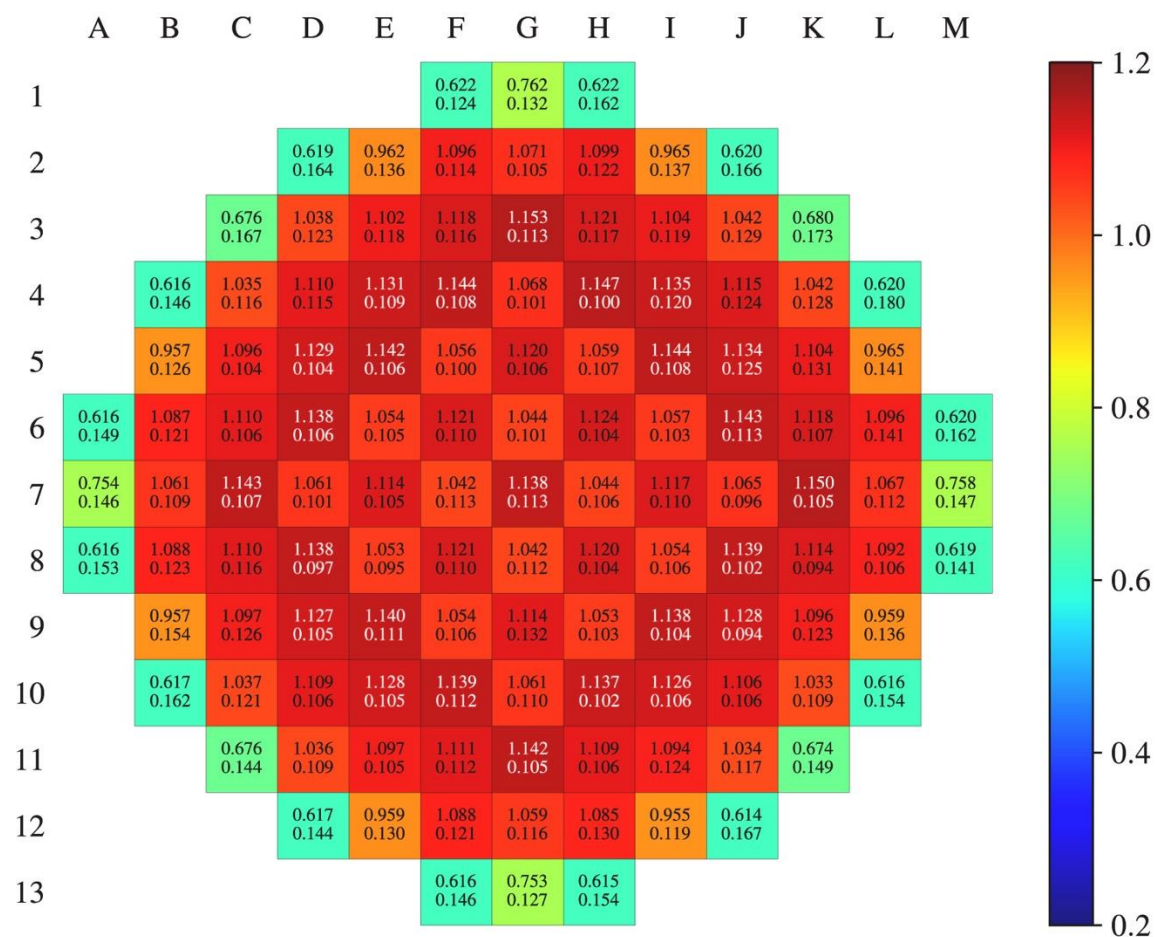
High accuracy

In a scenario where all 36 detectors were functional, the model was **highly precise**, with a **mean relative discrepancy** below **0.8%** and a **maximum discrepancy** under **2.5%** across all fuel depletion steps

Robustness to Failure

The researchers conducted a “stress test” by simulating the failure of individual detectors: even in the worst-case single-failure scenario, the **maximum relative discrepancy** remained low at **5.29%**, corresponding to those **assemblies** that **lacked symmetric counterparts** in other quadrants of the core

Results and interpretation



*Assembly-wise power distribution at BOC:
normalized relative power (above) and percent
relative uncertainty (below) for each fuel
assembly*

Overall, the GPR model successfully “learned” the complex power distribution **without requiring** pre-calculated “**calibration factors**” from diffusion simulations which can be less accurate than MC methods and unreliable if the reactor's actual state deviates from the simulation



Practical implications and key takeaways

07

Practical implications and key takeaways

Real-time monitoring

The GPR model is computationally efficient: it reconstructs power distributions for 10 depletion steps in just **7.15 seconds**, making it suitable for real-time monitoring of reactor safety margins

Reliable uncertainty estimates

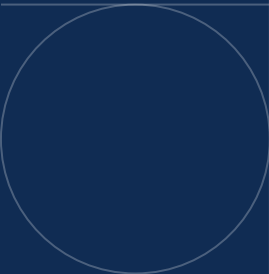
GPR provides a variance for every prediction, offering operators a built-in **measure of confidence**: this helps in identifying when the reconstruction might be less reliable due to sensor issues or unusual core conditions

Potential for anomaly detection

Since the method relies directly on detector data and **does not enforce a rigid parametric core model**, it can reveal unexpected anomalies (control rod drops, xenon spatial oscillations, fuel assembly failures) that traditional model-based approaches might overlook

Versatility

The study concludes that this methodology is **not limited to power mapping** and could be adapted for other critical tasks, such as reconstructing the core temperature distribution using data from fixed thermocouples



Thank you for the attention